Paleomagnetism of the Jan Lake Granite, Trans-Hudson Orogen[1]

M.G. Gala [2], D.T.A. Symons [3], and H.C. Palmer [2]

Gala, M.G., Symons, D.T.A., and Palmer, H.C. (1995): Paleomagnetism of the Jan Lake granite, Trans-Hudson Orogen; *in* Summary of Investigations 1995, Saskatchewan Geological Survey, Sask. Energy Mines, Misc. Rep. 95-4.

The Jan Lake granite refers to a complex of granite-pegmatite-aplite intrusions which are believed to be syn- to post-tectonic. Reported ages suggest that some elements of it are about 1770 Ma old. Stepwise thermal and alternating field demagnetization provide stable characteristic remanent magnetizations (ChRM) in 19 of 27 sites. The overall mean ChRM direction is D=126.3°, I=84.8°, α_{95}=7.7°, *k*=20, N=19 (pole: 48.1°N, 90.4°W, dp=15.1°, dm=15.3°). Because the ChRM directions define an elongated population, we have used Bingham (1974) statistics to estimate the extent of deviation of the data set from a true circular distribution. In fact, the data set divides into two distinct sub-populations (A and B) which, with the exception of three overlaps, fall under two separate geographic divisions. Using the fact that poles from the Trans-Hudson Orogen generally become younger from south to north, we are proposing that magnetization A (D=166.7°, I=73.2°, α_{95}=10°, *k*=27.6, N=10; pole: 24.3°N, 95.7°W) is an older magnetization which is probably older than 1844 Ma, the age of the Hanson Lake pluton magnetization. The B component (D = 026.3°, I = 80.3°, α_{95} = 6.8°, *k*=50.8, N=9; pole: 70.3°N, 78.2°W) is younger than 1844 Ma and probably about 1770 Ma old, the age obtained for undeformed late pegmatites in the Jan Lake granite complex and adjacent terranes.

Starting with the work of Dunsmore and Symons (1990), several other paleomagnetic studies have provided evidence that fit a collisional model for the Archean Slave-Rae-Hearn and Superior cratons (eg. Gala *et al.*, 1994; Symons, *et al.*, 1994). Estimates of the width of the Manikewan ocean (Stauffer, 1984), which was closed during this collision, are in excess of 4000 km. Recent paleomagnetic work in the Trans-Hudson Orogen has centred on the Hanson Lake Block. As part of this initiative we present results from a total of 272 specimens from 27 sites within the Jan Lake granite in the northern portion of the Hanson Lake Block.

1. Geology

The Hanson Lake Block is one of several lithotectonic domains within the Reindeer Zone of the Trans-Hudson Orogen (THO). These domains are believed to have been amalgamated during the closure of the Paleoproterozoic Manikewan ocean (Stauffer, 1984; Figure 1).

The Sturgeon-weir and the Tabbernor fault zones mark the block's eastern and western boundaries, respectively. It is separated from the Kisseynew Domain to the north by the sole zone of the overlying Kisseynew gneisses. Paleozoic sedimentary rocks lie on top of the Hanson Lake Block to the south (Figure 1).

The Pelican window is a domal core structure in the northern part of the Hanson Lake Block. The Sahli granite and the MacMillan granite are Archean inliers at the crest of the dome. Mantling the Archean inliers is a wide zone of synkinematic, highly strained rocks of amphibolite and granulite metamorphic grade (Figure 2; Ashton and Shi, 1994). Southward from the Sahli granite inlier and beyond the Jan Lake region by about 20 km to Tulabi Lake, the Hanson Lake Block is composed mostly of metamorphic supracrustal rocks (Macdonald and MacQuarrie, 1978). These rocks have been intruded by a granite-pegmatite-aplite complex that has

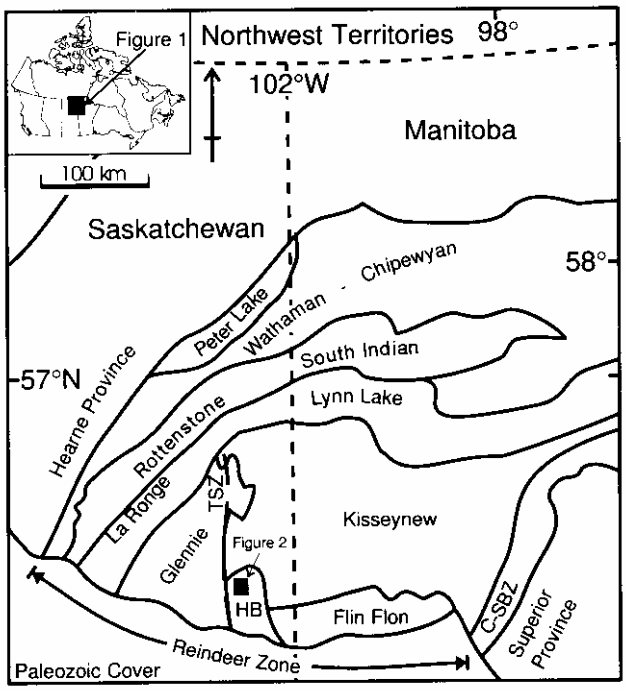


Figure 1 - Location of the study area and adjacent domains within the Reindeer Zone of the THO. Abbreviations: HB, Hanson Lake Block; TSZ, Tabbernor shear zone; and C-SBZ, Churchill-Superior boundary zone.

(1) Project funded by LITHOPROBE and NSERC operating grants; LITHOPROBE publication No. 697.
(2) Department of Earth Sciences, University of Western Ontario, London, ON N6A 5B7.
(3) Department of Earth Sciences, University of Windsor, Windsor, ON N9B 3P4.

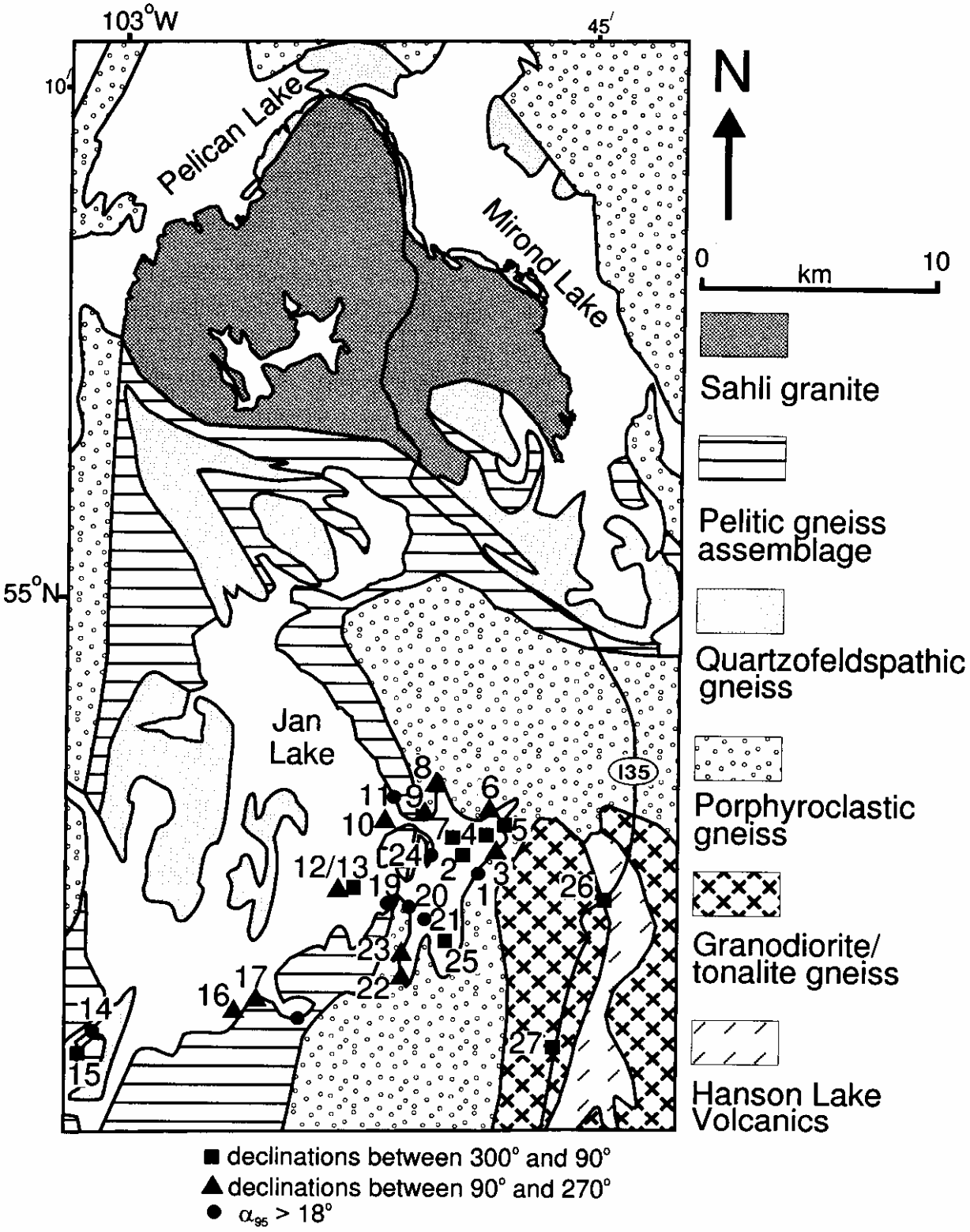


Figure 2 - Simplified geology of the study area showing the site locations in the Jan Lake granite. (■) and (▲) are sites that carry A and B components, respectively. (●) are sites with incoherent directions. Map adapted from Lewry et al. (1989).

been traditionally referred to as the Jan Lake granite and is believed to be syn- to post-tectonic. We shall keep this tradition and refer to the complex as a whole as the Jan Lake granite. The granite intrusion consists mainly of medium- to coarse-grained pink aplogranite and grey white leucogranite. Granite pegmatite occurs throughout much of the complex as unfoliated, thin sheets and hence post-dates major ductile deformation. A U-Pb zircon date on an aplitic-pegmatite gave 1773 ±9 Ma (Bickford *et al.*, 1987). Another U-Pb zircon age of 1767 ±1 Ma on undeformed pegmatite to the east of the Sturgeon-weir shear zone has been reported by Ashton *et al.* (1992). Also a Rb-Sr whole rock isochron of 1750 ±20 Ma has been obtained for the granite (Bell and Macdonald, 1982).

The peak of regional metamorphism in the adjacent Flin Flon and Kisseynew domains has been dated to be between 1815 and 1810 Ma (Gordon *et al.*, 1990). Recent results from radiometric age dating in the Pelican, Mirond, and Jan lakes area indicate that the peak of regional metamorphism occurred at about 1831 Ma (L. Heaman, pers. comm.).

2. Sampling and Experimental Work

Conventional paleomagnetic tools and methods were used to collect at least five oriented cores from each of 25 sites along the shore of Jan Lake and from two sites along Highway 135 (Figure 2). Up to three specimens, each 2.2 cm in length, were cut from each core giving a total of 272 specimens. Their remanent magnetizations were measured with a Canadian Thin Films Model DRM-420 cryogenic magnetometer in a magnetically-shielded room. A total of 59 specimens were alternating field (AF) demagnetized in 11 steps up to 110 mT using a Sapphire Instruments SI-4 demagnetizer. Another 32 specimens were stepwise thermally demagnetized in up to 11 steps up to 680°C using a Magnetic Measurements MM1 demagnetizer. A further 84 specimens were AF demagnetized in four or five steps up to 90 mT. The remaining 97 specimens were initially AF demagnetized at 30 or 30 and 50 mT, and then thermally demagnetized in two or three steps up to 570°C.

Constant temperature acquisition magnetization curves were determined for 17 test specimens using up to 11 steps up to 900 mT in order to study their saturation isothermal remanent magnetization (SIRM) characteristics. The corresponding decay curves were obtained by AF demagnetization of the acquired SIRM using six steps up to 80 mT.

Characteristic remanent magnetization (ChRM) directions were obtained using the least-squares multivariate method of Kirschvink (1980). Accepted ChRM directions had maximum angular deviations of less than 10°. The remagnetization circle method (Bailey and Halls, 1984) was used to obtain the ChRM of a few specimens. Site means were computed using Fisher (1953) statistics. The unit mean has been calculated using both Fisher (1953) and Bingham (1974) statistics.

3. Results

The natural remanent magnetization intensities of the specimens are in the order of 10^{-1} to 10^{3} mA/m. The ChRM directions were usually defined between 10 and 90 mT for AF and between 150° and 580°C for thermal demagnetization. The intensity of ChRM ranged from about 10^{-1} to 10^{3} mA/m with most specimens between 10^{-1} to 10 mA/m. Examples of specimens with low coercivity are shown on Figures 3a and 3b. Some of the specimens with high coercivity (Figure 3c) were thermally demagnetized (Figure 3d).

Thermal demagnetization in a majority of the specimens reveals the presence of both pyrrhotite and magnetite (Figures 4c and 4d). Pure magnetite and titanomagnetite with possibly some hematite are the magnetic carriers in a minority of the specimens (Figure 4a and 4b). Figures 4a and 4d also show that the conversion of sulphides to magnetite has taken place in some specimens at high-demagnetization temperatures.

The SIRM and corresponding decay curves (Figures 5a and 5b) indicate that the dominant magnetic carriers are multidomain to single domain magnetite (or pyrrhotite because magnetite and pyrrhotite characteristics are similar). However, the specimen of pegmatite at site

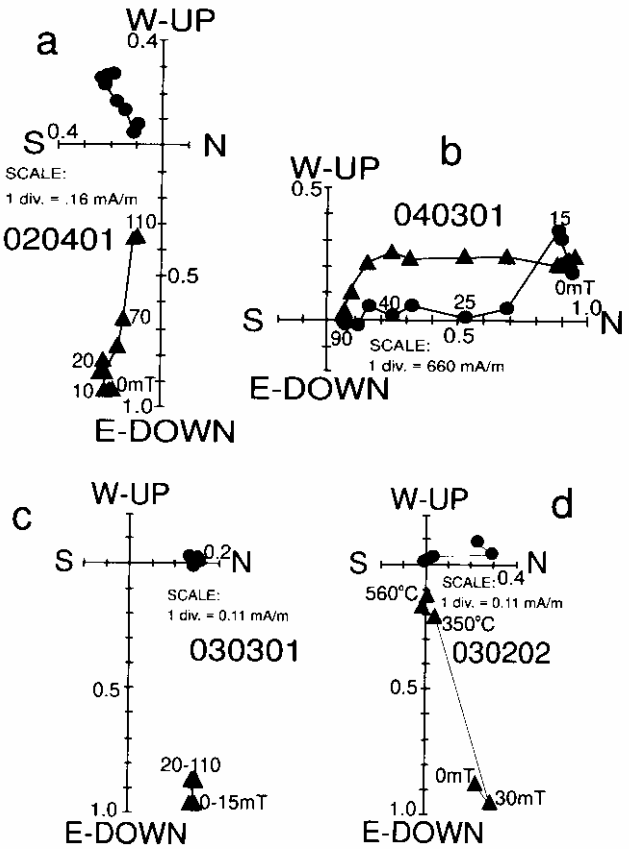


Figure 3 - Orthogonal vector diagrams for representative specimens subjected to alternating-field demagnetization. Circles and triangles are points on the horizontal and vertical planes, respectively. The six character numbers denote the site, core, and specimen numbers of two characters each.

12 illustrates fine-grained hematite characteristics, and the gneiss specimen at site 5 illustrates coarse-grained hematite behaviour.

The site means and their associated statistical parameters are given in Table 1. Eight sites have means with α_{95} values >18° and so they have not been used to calculate the unit mean. Of the remaining 19 site means, 12 are defined with α_{95} <12°. The Fisher (1953) and Bingham (1974) means of the 19 sites are represented in Table 1 by M_f and M_{bh}, respectively. In Figure 6, we plot the site means on the lower hemisphere of an equal-area projection.

4. Discussion

The Jan Lake granite overall mean ChRM direction gives a pole at 48.1°N and 90.4°W (dp=15.1°, dm=15.3°). This places it at a much higher paleolatitude than the 1855 ±6 Ma pole for the Wathaman Batholith (WB, Figure 7; Symons, 1991), the 1869 Ma pole for the Peter Lake Domain (PD, Figure 7; Symons, 1994), and the 1855 Ma composite pole for the Slave-Rae-Hearne cratons (SRH, Figure 7; Symons, 1991). However, the Jan Lake granite pole's high paleolatitude is consistent with other results from the Hanson Lake Block (HP, SG, Figure 7; Gala *et al.*, 1994; Gala *et al.*, in prep.) and from other domains within the Reindeer Zone of the THO (RE, BP, MC, Figure 7; D.T.A Symons, unpubl. data; Mackay, 1992; Symons *et al.*, 1994). Thus this Jan Lake granite pole adds to the growing evidence that the "north-south" width of the Paleoproterozoic Manikewan ocean was in excess of 4000 km, with its boundary at the southeast side of the Wathaman Batholith (Symons *et al.*, in press).

Two other paleomagnetic results from the Hanson Lake Block have been obtained. One is from the massive 1844 ±2 Ma Hanson Lake pluton which has intruded the Hanson Lake metavolcanics in the southeastern part of the Hanson Lake Block. It is exposed over an area of 50 km^2 and it ranges in composition from pyroxenite to granodiorite. The other is the reworked Archean Sahli granite which is exposed in a window in

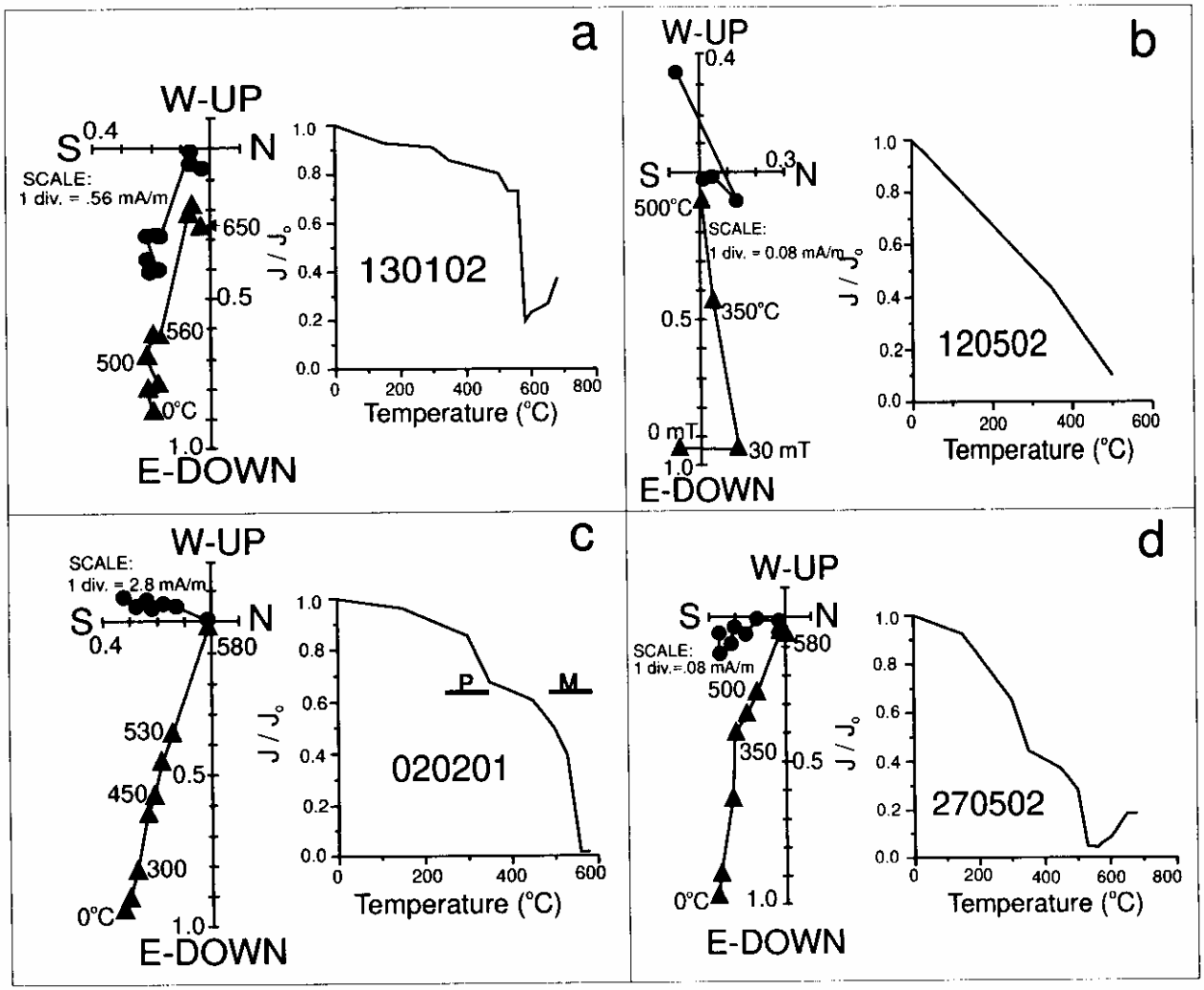


Figure 4 - Orthogonal vector and corresponding normalized intensity diagrams for representative thermally-demagnetized specimens. Symbols are as defined in Figure 3.

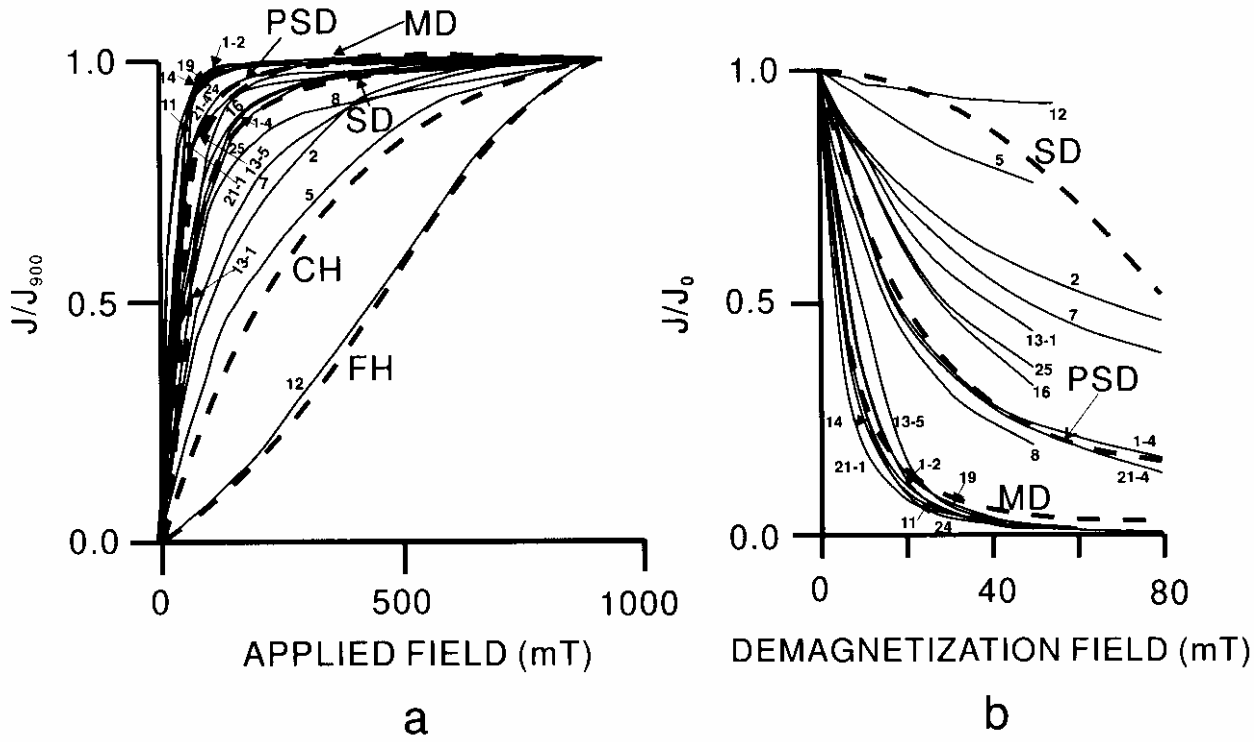


Figure 5 - a) Saturation isothermal remanent magnetization (SIRM) acquisition curves for 17 representative specimens and b) Corresponding alternating field demagnetization curves. Abbreviations: SD, PSD, and MD, single domain, pseudosingle domain, and multidomain magnetite; CH and FH, coarse- and fine-grained hematite.

the northern part of the Hanson Lake Block. Overlying the Sahli granite are gneisses which have been intruded by the Jan Lake granite. The Sahli granite is exposed over an area of 100 km^2 and it is composed of coarse-grained and variably-retrogressed granulite-facies gneissic monzocharnockite (Lewry *et al.*, 1989). Regional metamorphism in this area probably peaked at about 1831 Ma (L. Heaman, pers. comm.).

The Hanson Lake pluton and Sahli granite poles are at, 36.2°N and 94.5°W and 49.2°N and 84.0°W, respectively. Thus the Jan Lake granite pole is somewhat between the Hanson Lake pluton and Sahli granite poles. In fact, the error circles around the Sahli granite and Jan Lake granite poles overlap. Paleomagnetic evidence suggests that the Sahli granite ChRM records the metamorphic event which probably peaked around 1831 Ma (Gala *et al.*, in prep.; L. Heaman, pers. comm.). If this is true, then the agreement between the Sahli granite and Jan Lake granite poles suggests that the Jan Lake granite magnetization was acquired during this metamorphic event. This is in conflict with two ages (1773 ±9 Ma, Bickford *et al.* 1987; 1767 ±1 Ma, Ashton *et al.*, 1992) obtained for undeformed pegmatite in the Jan Lake granite complex. Moreover, the paleomagnetic evidence notwithstanding, recent tectonic models suggest that the Sahli granite existed as part of a separate Archean microcontinent which was underthrust, from the southwest, below a coherent Glennie–Hanson Lake–Flin Flon protocontinent by 1830 Ma. The resolution of this conflict may lie in the fact that the Jan Lake granite complex consists of multiple intrusions which differ significantly in age. We suggest that the moderately foliated granite intrusion, which contributed some of our samples, is older than 1831 Ma and the massive unfoliated pegmatites that were radiometrically dated are about 1770 Ma old. The paleomagnetic evidence for this proposal lies in the fact that the Jan Lake granite ChRM directions define an elongated population (Figure 6).

Bingham (1974) statistical method provide a unit mean of D=126.3°, I=84.8°, semi-minor axis $\alpha_{1\text{-}3}$=5.4°, semi-major axis $\alpha_{1\text{-}2}$=8.6°, N=19 (Table 1, Figure 6). The Bingham statistics, in addition to providing a measure of the circular dispersion as predicted by Fisher statistics, also provides a measure of the deviation of the data from a true circular distribution. Thus the semi-major axis ($\alpha_{1\text{-}2}$) includes estimates of the circular dispersion and the bias whereas the semi-minor axis ($\alpha_{1\text{-}3}$) gives an estimate of the circular dispersion only. Although the Bingham statistics' mean does not differ from the Fisher statistics' mean, the implications of the elongated population could be far reaching.

First, since the sampling was not restricted to only the moderately foliated granite intrusion, part of the elongation could arise from the presence of older and/or younger magnetizations in moderately foliated and unfoliated granites, respectively. In support of this possibility are the results from between sites 12 and 13 (Table 1, Figure 6). Site 12 is in a pegmatite sheet which is in contact with the moderately foliated granite of site 13. Taken alone, the two site mean ChRM directions are

Table 1 - Site mean ChRM directions.

Site	Geol.	Number of specimens				Demagnetizing range		Mean remanence vector			
		a	b	c	d	mT	°C	Decl. °	Incl. °	α_{95} °	*k*
1*	g	5	11	3	05	10-90	300-350	322.0	66.0	48.3	04
2	g	5	12	5	06	10-110	150-580	246.5	85.3	12.0	32
3	g	5	09	5	09	00-110	350-560	351.9	79.0	05.1	102
4	g	5	10	3	05	25-90	150-450	164.3	48.3	06.7	132
5	gn	5	10	5	10	10-110	500-580	168.5	65.3	10.0	25
6	g	5	10	4	07	10-110	150-570	081.7	81.2	16.2	15
7	g	5	12	3	05	30-90	350-560	125.1	64.4	12.8	37
8	g	5	11	5	06	10-90	150-450	339.3	72.6	16.5	17
9	g	5	09	5	06	00-50	300-560	317.4	86.6	09.2	55
10	g	5	11	5	10	10-110	350-580	335.8	81.9	09.0	30
11*	g	5	11	4	05	30-70	300-450	295.5	62.9	61.1	03
12	p	5	10	5	10	10-110	150-560	013.3	77.0	07.5	42
13	gn	5	11	4	08	05-110	530-680	163.1	78.4	15.7	14
14*	g	5	12	5	09	10-110	150-350	128.2	65.5	24.0	07
15	g	5	10	3	05	10-110	150-550	239.0	69.5	11.3	47
16	g	5	09	5	08	20-110	150-550	087.2	62.5	10.4	29
17	g	5	09	3	04	10-90	30mT-550	027.2	68.4	04.5	381
18*	g	5	08	4	04	10-110		055.1	71.5	33.4	09
19*	g	5	10	5	09	15-110	150-300	034.5	57.6	26.7	05
20*	g	5	10	3	04	10-90		098.5	67.4	23.3	17
21*	g	5	08	4	05	10-110		199.0	66.9	48.0	04
22	g	5	09	4	06	10-110	30mT-350	033.3	80.5	10.1	45
23	g	5	10	4	04	10-70	30mT-350	019.2	82.9	17.5	29
24*	g	5	11	3	05	10-90		324.5	75.4	41.3	05
25	g	5	11	4	08	10-110	30mT-580	119.4	78.7	07.5	56
26	g	5	08	5	08	05-110	30mT-500	174.4	69.7	15.3	14
27	g	5	10	4	08	10-110	350-560	166.6	71.9	07.7	53
M_f					(19)			126.3	84.8	07.7	20
M_{bh}					(19)			126.3	84.8	05.4	9.6

Notes:
* Site means with an α_{95} of greater than 18°; these were not used to compute the unit mean. "Geol." is the type of rock sampled (g=granite, gn=gneiss, p=pegmatite sheet). "a" and "b" record the number of cores and specimens measured, and "c" and "d" represent the number of cores and specimens used to obtain the site mean. "Decl.", "Incl.", α_{95}, *k* are declination, inclination, cone of 95 percent confidence limit, and precision parameter (Fisher, 1953), respectively. "M_f" and "M_{bh}" are the unit means using Fisher (1953) and Bingham (1974) statistics, respectively. The values recorded under α_{95} and *k* for "M_{bh}" are the minor and major axes of the Bingham error ellipsoid.

significantly different implying that they represent magnetizations of different ages from two different units.

Second, metamorphism and/or co-incident intrusion probably took place over a sufficiently long time or in discrete stages for apparent polar wander (APW) to occur. Thus, different sections of the Jan Lake granite complex cooled at different times and hence acquired different magnetizations. The best support for this model would be for the elongated population to correlate with some geographic and/or geologic distribution of the sites. Although we fail to see any division based on rock type, we see a weak geographic division of the population into two major groups; one is dominated by sites to the west having remanence declinations between 90° and 270° (A component) and the other is dominated by sites to the east having remanence declinations between 300° and 90° (B component) (Figures 2 and 6). The means of these groupings are A: D=166.7°, I=73.2°, α_{95}=10°, *k*=27.6, N=10; and B: D=026.3°, I=80.3°, α_{95}=6.8°, *k*=50.8, N=9 with paleomagnetic poles at 24.3°N, 95.7°W and 70.3°N, 78.2°W, respectively. The two means, which are both defined with better precision than the combined mean, are statistically different and may be considered to represent different magnetizations of different ages. Symons *et al.* (in press) have deduced that the THO poles generally become younger in a south to north direction. These means follow this trend with the A component being the older of the two; perhaps older than 1844 which is the age of magnetization of the Hanson Lake pluton (Figure 7). The B component is then younger than 1844 Ma but probably records the reported age of about 1770 Ma for undeformed pegmatites within the Jan Lake granite.

5. Comments

New paleomagnetic results from the 1775 Ma Deschambault Lake pegmatites provide a paleolatitude that is almost identical to the B component paleolatitude. If it is true that these pegmatites belong to the group of post-tectonic pegmatites that extend from the La Ronge belt to the Churchill-Superior boundary, then the Deschambault pegmatites result supports our proposal that the B component is about 1770 Ma old.

In addition, the 1770 Ma poles are close to the 1838 Ma Boot Lake–Phantom Lake pole. Whether this

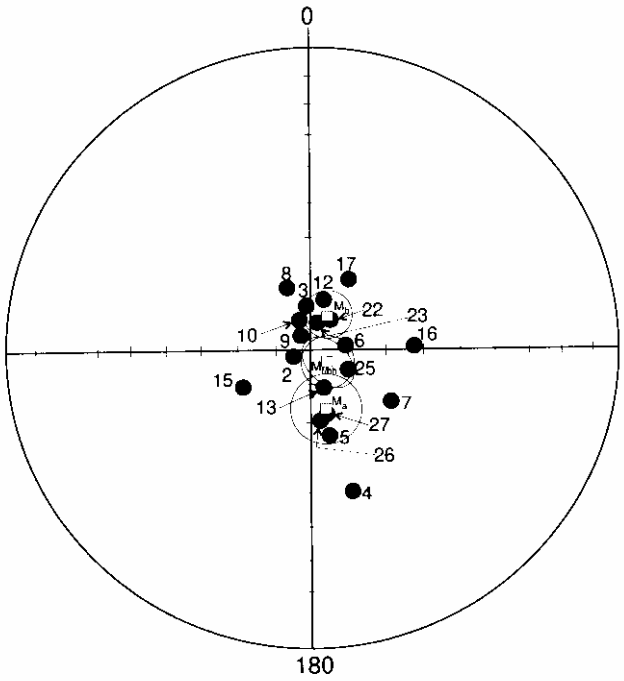


Figure 6 - Lower-hemisphere, equal-area projection of the site mean ChRM directions. The Fisher and Bingham statistics for the overall means for sites with $\alpha_{95}<18°$ are represented by M_f and M_{bh}, respectively. They are encircled by their cone and oval of 95 percent confidence. Similarly the A and B components Fisher means are represented by M_a and M_b, respectively.

is because the geomagnetic field did not change significantly between 1838 and 1770 Ma or due to relative tectonic motion between terranes within the THO will be addressed when we present the details of the Sahli granite results.

In the above discussion we have not attempted to tilt-correct all the three paleomagnetic results from the Hanson Lake Block. A common problem usually associated with plutonic rocks as opposed to volcanic rocks, is that it is not easy to define a paleohorizontal for them. This renders any tilt-correction problematic and ambiguous. However, the Pelican window is a very gentle and round domal structure. This is probably evidence that the Sahli and the Jan Lake granites could not have been substantially tilted. We therefore, at least for now, believe that these magnetizations have not been significantly tilted.

6. Acknowledgments

We would like to thank Ken Ashton for geologic advice, and Hairu Pan and Brock Symons for field assistance. Brock Symons also assisted in specimen preparation. Jaros Kulhanek assisted in measuring. One of the authors, MGG, is a Canadian Commonwealth Scholar and is on study leave from Federal University of Technology, Yola, Nigeria. This work was supported financially through NSERC and LITHOPROBE operating grants to DTAS.

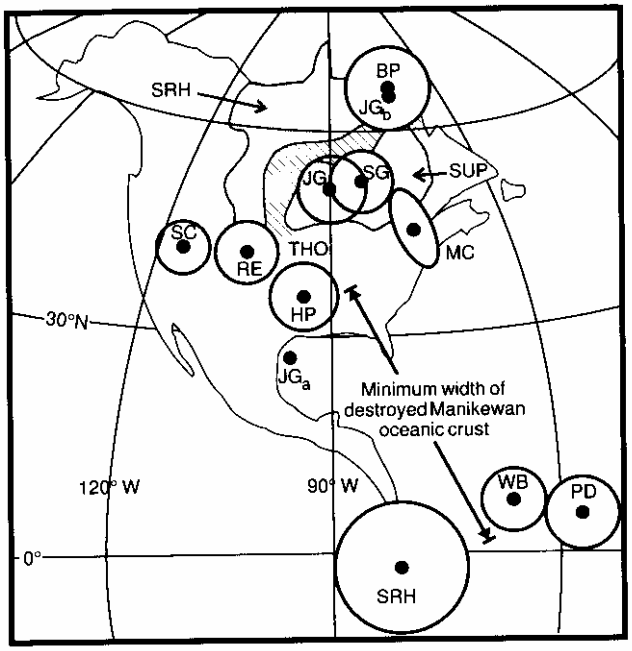


Figure 7 - Poles from the Trans-Hudson Orogen (THO) are Macoun Lake pluton (MC, Lynn Lake–La Ronge Domain), Hanson Lake pluton (HP, Hanson Lake Block), Jan Lake granite (JG, JG_a, JG_b, Hanson Lake Block), Sahli granite (SG, Hanson Lake Block), Reynard Lake pluton (RE, Flin Flon Domain), Boot Lake–Phantom Lake pluton (BP, Flin Flon Domain), Peter Lake Domain (PD), and Wathaman Batholith (WB). The Slave-Rae-Hearne (SRH) and Superior (SUP) cratons are represented by the composite SRH and Sudbury complex (SC) poles, respectively.

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